

Electrical, structural, and photocatalytic properties of copper oxide nanowire

Azadeh Jafari¹ , SAA Terohid², Alireza Kokabi³
and Amir Moradiani⁴

Abstract

Using a direct oxidation method in a horizontal quartz tube, copper oxide nanowires are grown on a Cu substrate. In order to investigate the growth temperature effects on the structural, morphological, electrical, and photocatalytic properties of the copper oxide nanowires, X-ray diffraction, scanning electron microscopy, a KEITHLEY 2361 system, and a homemade photoreactor are used. The X-ray diffraction results show that both CuO and Cu₂O phases are formed, and while increasing the growth temperature, the crystallinity is improved and the intensity of most of the diffraction peaks increases. The scanning electron microscopy images at different growth temperatures show that the number, density, and length of the copper oxide nanowires on pre-formed micro-scaled grains increase, when the growth temperature increases to 700°C and sharper nanowires with average diameters of 1–3 µm grow on the surface. Also I–V curves show that by raising the growth temperature, the conductivity of the samples increases. In addition, the photocatalytic activities are studied by photocatalytic degradation of Congo red dye, and based on these results, the sample grown at 700°C with the highest number and density of the nanowires showed the best photocatalytic performance and electrical conductivity. The results can be used to guide better understanding of the growth behavior of copper oxide nanowires and can be useful for the development of novel photocatalytic nanodevices.

Keywords

copper oxide nanowires, electrical properties, morphology, photocatalytic activity, structure

Date received: 30 June 2019; accepted: 11 December 2019

Introduction

Copper oxide has been considered as a p-type metal–oxide semiconductor among the metal oxides.^{1–11} Most recently, one-dimensional structures, copper oxide and particularly copper oxide nanowires (NWs), have received extensive attention because of their interesting properties.^{4,5} They are used in field-effect transistors, photovoltaic cells, photo-thermal and photoconductive materials, field emission nanodevices, and chemical and gas sensors.^{8,10}

CuO has been grown in a variety of nano-scale forms, including as nanoparticles, nanofilms, nano-flowers, nanorods, nanofibers, nanoribbons, nanoparticles, and NWs.^{11–15} It is of interest to explore methods for synthesizing high-quality CuO NWs with low cost and in large quantities. Various CuO NW synthetic methods have been reported including hydrothermal, electrochemical, polyol, template-based sol–gel, and direct oxidation of copper or

copper-containing materials.^{11–17} Unlike direct oxidation approaches, these other synthetic methods may result in impurities in the NWs.^{16,17} Recently, the direct oxidation of various types of copper substrates such as sheets, foils, and grids is the simplest and most suitable method for the large-scale and vertically aligned production of copper oxide NWs.^{18,19} It has been reported that copper oxide NWs are suitable as photoresists of some colors and also the photocatalytic performance of copper oxide NWs is another interesting issue which has been reported rarely, but because of the importance of environmental pollutants, it needs to be studied more. In this work, we report the successful fabrication of vertically aligned copper oxide NWs on Cu foil using thermal oxidation in air at different growth temperatures. Samples were characterized by X-ray diffraction (XRD), scanning electron microscopy (SEM), with a KEITHLEY 2361 system and using a homemade photoreactor in order to

¹South Ural State University, Chelyabinsk, Russia

²Department of Physics, Islamic Azad University, Hamedan, Iran

³Department of Electrical Engineering, Hamedan University of Technology, Hamedan, Iran

⁴Department of Materials Engineering, Tarbiat Modares University, Tehran, Iran

Corresponding author:

Azadeh Jafari, South Ural State University, Chelyabinsk 454080, Russia.

Email: jafari_manesh@yahoo.com

investigate the structural, morphological, electrical, and photocatalytic properties of copper oxide NWs.

Results and discussions

XRD analyses

The XRD patterns of the three samples at growth temperatures of 500°C, 600°C, and 700°C are shown in Figure 1. By heating the copper foil and interaction of oxygen with the copper atoms, two different chemical structures could be created in the form of CuO and Cu₂O, and according to the XRD patterns, there are mixed phases of CuO and Cu₂O in all the samples.

Another noteworthy point is that the intensity of most of the diffraction peaks increased in increasing the growth temperature and no new peaks were observed. In other words, thermal energy caused by the high temperature leads to enhancement of the mobility of the active sites on the copper foil surface. The increase of mobility can be attributed to grain growth and the reduction of defects in increasing the temperature. On the other hand, according to Scherrer's formula, the grain size decreases. Considering the XRD results, there is a possibility that the NWs have grown on the surface of a CuO/Cu₂O/Cu multilayer structure. The chemical reactions to synthesize the copper oxide NWs in both CuO and Cu₂O phases can be described by the following two equations:²⁰

1. $4\text{Cu} + \text{O}_2 = 2\text{Cu}_2\text{O}$;
2. $2\text{Cu}_2\text{O} + \text{O}_2 = 4\text{CuO}$.

As a result, the Cu₂O layer is grown first, and then the CuO layer is grown on the Cu₂O layer.

SEM analysis

Figure 2 shows the SEM images for the surface of the Cu foils after oxidation for 5 h at 500°C, 600°C, and 700°C. As can be seen from the image of sample M1, at the first stage of growth process with the oxidation of the Cu foil, a thin film of copper oxide is created in the form of micrograins with continuous and dense cluster shapes. Gradually, with the passage of time, NWs with nearly curved shapes begin to germinate on micrograins; therefore, NW with small diameter is formed and continues to grow, as shown in Figure 2.

Figure 2 shows that as the growth temperature increases, the density of the NWs, increases the coverage on the Cu substrate improves and the NWs become denser. When the growth temperature increases to 700°C (sample M3), more sharp NWs with average diameters of 1–3 µm grow on the surface. It can be concluded that the growth temperature has an important effect on the length and density of NWs because of compressive stress, which creates multilayer structure on the CuO/Cu₂O/Cu. By diffusion of Cu ions across grain boundaries in the copper oxide layers and from bulk copper to the surface, copper oxide NWs grow on the copper oxide layer surface. In fact, copper oxide micrograins act as a template for initiating nucleation and the growth of copper oxide NWs.^{13–16}

To explain the diameter dependence and temperature dependence of growth direction and the length of the NWs, several models based on the energy analysis of grown NWs estimating the surface and interface energies of the NWs have been proposed. These models calculate the total free energy of NWs with different growth directions, and the most favorable growth direction with the lowest free energy can be found. The parameter is usually estimated to approximately fit the experimental observations and is considered to be a constant for different NW diameters. It is as yet unclear what determines the value

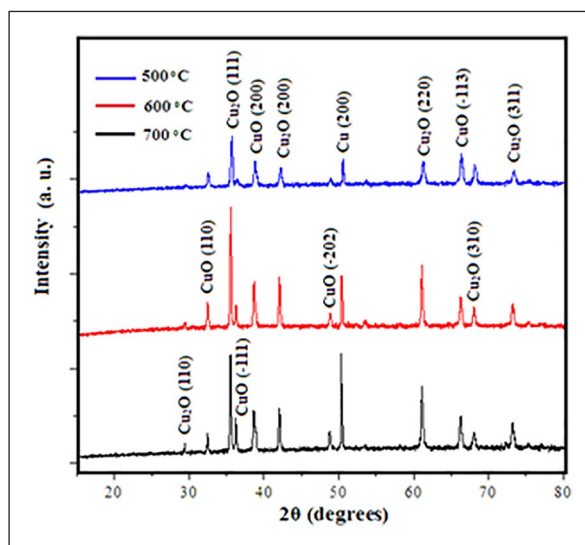


Figure 1. XRD patterns of the samples prepared at 500°C, 600°C, and 700°C.

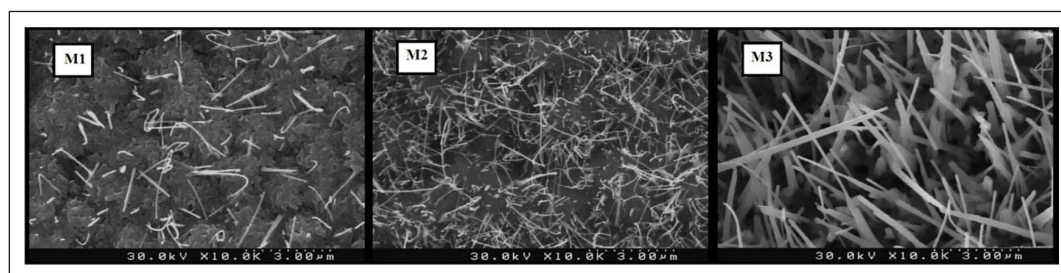


Figure 2. SEM images of the samples: M1, M2, and M3 prepared at temperatures of 500°C, 600°C, and 700°C.

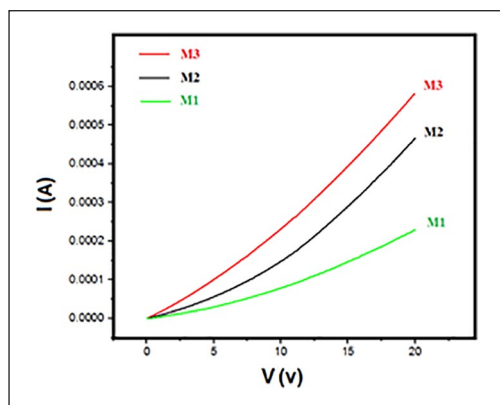


Figure 3. Dependence of the current and voltage of the samples. M1, M2, and M3 prepared at various temperatures of 500°C, 600°C, and 700°C.

of the thickness. In fact, the critical thickness in the calculation of NW energy may be influenced not only by growth temperature, but also by NW diameter, because the initial growth stage is also dependent on the diameter.^{19–21}

Electrical properties

Figure 3 illustrates the current as a function of voltage from 0 to 20 V determined with a KEITHLEY 2361 system. A linear behavior that can be seen from I–V curves is due to the metallic electrical property of the copper oxide NWs. It has been reported that in semiconductor NWs, hopping conductivity across surface and interface states leads to electrical conductance. Since the ratio of the surface to the volume in copper oxide NWs is a significant amount, also by increasing the growth temperature, the conductivity of the samples increases. This may be due to an increase in the number of NWs on the sample surface which increases the available free charge carriers and conduction channels.²¹ On the other hand, it was found that the number and size of the NWs have a clear influence on the electrical properties, which is in good agreement with theoretical models.^{18,19} According to the Matthiessen formula, the conductivity can be explained based on various electron-scattering mechanisms due to impurities, defects, grain boundaries, and surface scattering in thin films,²² and in this case, increasing the growth temperature results in an increased number of grain boundaries.

Photocatalytic property

Figure 4 illustrates the degradation of Congo red by samples of the copper oxide NWs at different temperatures. In this case, C0 is the initial dye concentration and Ce is the concentration of the dye after photodegradation (mg L^{-1}). As can be seen from Figure 4, the Ce/C0 values decrease rapidly during the first hour of irradiation.^{23,24} During the photocatalysis process when the photon energy of the ultraviolet (UV) source is larger than or equal to the band gap of the copper oxide NWs, electrons in the valence band can be excited to the conduction band and photo-induced electron–hole pairs were created on the surface of

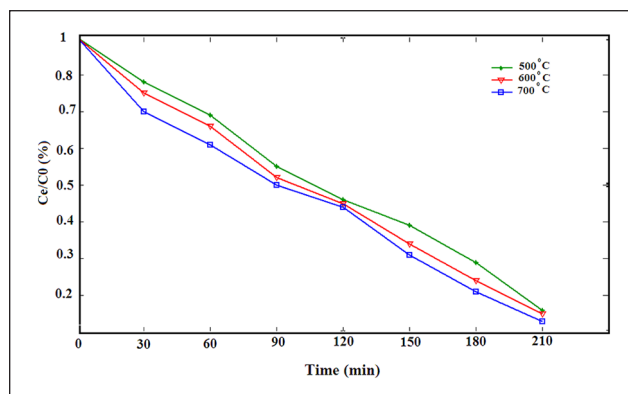


Figure 4. Photocatalytic degradation of Congo red versus reaction time for nanowires prepared at temperatures of 500°C, 600°C, and 700°C.

the copper oxide NWs. It is known that the photocatalytic activity of the catalyst depends on its ability to reduce and oxidize photoelectrons and holes generated under irradiation.^{23–25} Also, it can be observed that the NWs grow at 700°C have better photocatalytic performance, which is due to their higher density, length, and diameter, as can be seen from the SEM images.

Conclusion

In this study, the direct oxidation method to grow copper oxide NWs has been used. The effect of the growth temperature on the structural, morphological, and electrical properties of the copper oxide NWs has been investigated. Based on XRD results, it was found that as the growth temperature was increased, the crystallinity improved gradually and CuO and Cu₂O phases appeared. Surface morphologies measured by field emission scanning electron microscopy (FESEM) revealed that gradually, the number and the length of the NWs on the copper oxide grains increased when the growth temperature increased. Also, electrical measurements showed that when the growth temperature was increased from 500°C to 700°C, the conductivity increased due to an increase in the number of NWs on the sample surface which increases the number of available free charge carriers. The NW grown at 700°C was the longest, and had the highest conductivity, a well-crystallized structure, and demonstrate better performance in degradation of Congo red dye, possibly due to a higher NW density and aspect ratio.

Experimental procedure

Copper oxide NW was prepared by thermal oxidation in a horizontal quartz tube in the presence of air. Thus, a 1 cm × 1 cm copper foil (99.9% purity from the Merck Germany) with a thickness of 0.15 mm was used as the substrate and starting material. Before the thermal oxidation process, using ethanol and acetone (from Merck Germany) and double distilled water, the copper foils were cleaned ultrasonically. Afterward, the Cu foils were placed in the center of the tube and heated to the desired

temperatures of 50°C, 600°C, and 700°C to prepare three series of samples: M1, M2, and M3. In order to control the growth temperature, the air-oven temperature was monitored continuously by placing a thermocouple in the vicinity of the Cu foil. After a reaction time of 5 h, the Cu foil samples were oxidized and finally, a black layer was formed on the Cu foil surface. To avoid thermal shocks and to achieve a crack-free NW, the samples were slowly cooled to room temperature. Several analytical techniques were applied to characterize the samples. The morphology of the samples was characterized by SEM (Leo 440i) and the crystal structure and the phase composition were analyzed by XRD using CuK α radiation. Also in this experiment, the current–voltage characteristic was studied with a KEITHLEY 2361. In addition, the dye powder was dissolved in deionized water with a concentration of 20 mg L⁻¹ in order to prepare a solution for investigation of the photocatalytic activity. Next, 0.1 g of the copper oxide NWs was added to 100 mL of dye solution. Afterward, the solutions were placed in a homemade photoreactor and illuminated under a UV source.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iD

Azadeh Jafari  <https://orcid.org/0000-0002-5765-5628>

References

1. Wang C, Fu XQ, Xue XY, et al. *Nanotechnology* 2007; 18: 5.
2. Liu L, Sheng YY, Liu M, et al. *Tribol Int* 2019; 140: 105727.
3. Gou XL, Wang GX, Yang JS, et al. *J Mater Chem* 2008; 18: 965.
4. Zhu YW, Yu T, Cheong FC, et al. *Nanotechnology* 2005; 16: 88.
5. Dastan D, Gosavi SW and Chaure NB. *Macromol Symp* 2015; 347: 81.
6. Zheng XG, Xu CN, Tomokiyo Y, et al. *Phys Rev Lett* 2000; 85: 5170.
7. Dastan D and Chaure NB. *J Mater Mech Manufact* 2014; 2: 21.
8. Anandan S, Wen XG and Yang SH. *Mater Chem Phys* 2005; 93: 35.
9. Dastan D, Chaure N and Kartha M. *J Mater Sci: Mater Electron* 2017; 28: 7784.
10. Husain A, Hone J, Postma HWC, et al. *App Phys Lett* 2003; 83: 1240.
11. Dastan D, Panahi SL and Chaure NB. *J Mater Sci: Mater Electron* 2016; 27: 12291.
12. Shinde SK, Dubal DP, Ghodake GS, et al. *Nano-Structures Nano-Objects* 2016; 6: 5.
13. Zhu X, Yang J, Dastan D, et al. *Composites Part A* 2019; 125: 105521.
14. Dastan D. *Appl Phys A* 2017; 123: 13.
15. Martinson CA and Reddy KJ. *J Colloid Interface Sci* 2009; 336: 406.
16. Dastan D. *J Atomic, Molecul. Condensate Nano Phys* 2015; 2: 109.
17. Gao XP, Bao JL, Pan GL, et al. *J Phys Chem B* 2004; 108: 5547.
18. Panahi SL, Dastan D and Chaure NB. *Adv Sci Lett* 2016; 22: 941.
19. Gonçalves AMB, Campos LC, Ferlauro AS, et al. *J App Phys* 2009; 106: 034303.
20. Dastan D. *Appl Phys A* 2017; 123: 13.
21. Huang L, Li D, Chang P, et al. *Phys Rev B* 2011; 83: 245310.
22. Werninghaus T, Hahn J, Richter F, et al. *Appl Phys Lett* 1997; 70: 958.
23. Sudha D and Sivakumar P. *Chem Eng Process: Process Inten* 2015; 97: 112.
24. Yu Z, Chen XQ, Kang X, et al. *Adv Mater* 2018; 30: 1706259.
25. Pore V, Heikkilä M, Ritala M, et al. *Photochem Photobiol A* 2006; 177: 68.